simultaneously live more fully in the moment. Maybe this can be the meaning we seek.

We're definitely getting closer to an answer. Whether or not the world is at any given moment falling apart from a political perspective, scientifically we are living in a golden age. In physics, recent discoveries and new technological and theoretical tools are allowing us to make leaps that were previously impossible. We've been refining our understanding of the beginning of the universe for decades, but the scientific exploration of how the universe might end is just now undergoing its renaissance. Hot-off-the-presses results from powerful telescopes and particle colliders have suggested exciting (if terrifying) new possibilities and changed our perspective on what is likely, or not, in the far future evolution of the cosmos. This is a field in which incredible progress is being made, giving us the opportunity to stand at the very edge of the abyss and peer into the ultimate darkness. Except, you know, quantifiably.

As a discipline within physics, the study of cosmology isn't really about finding meaning per se, but it is about uncovering fundamental truths. By precisely measuring the shape of the universe, the distribution of matter and energy within it, and the forces that govern its evolution, we find hints about the deeper structure of reality. We might tend to associate leaps forward in physics with experiments in laboratories, but much of what we know about the fundamental laws governing the natural world comes not from the experiments themselves, but from understanding their relationship to observations of the heavens. Determining the structure of the atom, for example, required physicists to connect the results of radioactivity experiments with the patterns of spectral lines in the light from the Sun. The Law of Universal Gravitation, developed by Newton, posited that the same force that makes a block slide down an inclined plane keeps the Moon and planets in their orbits. This led, ultimately, to Einstein's General Theory of Relativity, a spectacular reworking of gravity, whose validity was confirmed not by measurements on Earth, but by observations of Mercury's orbital quirks and the apparent positions of stars during a total solar eclipse.

Today, we are finding that the particle physics models we've developed through decades of rigorous testing in the best Earthly laboratories are incomplete, and we're getting these clues from the sky. Studying the motions and distributions of other galaxies—cosmic conglomerations like our own Milky Way that contain billions or trillions of stars—has pointed